SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE:ITA0518**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

Code of Conduct:

I PRBHAVATHI R(192321101) Certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
PRABHAVATHI R

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.

(b) Apply a negative transformation $s = 255 - r$ to the pixel values:

$r = [50,100,150,200]$ Compute the transformed values.

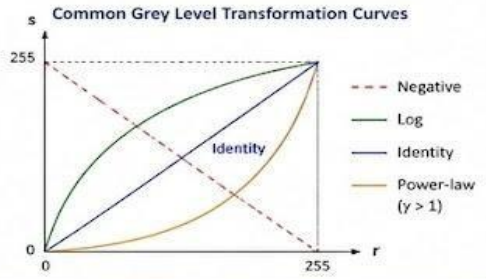
1. GREY LEVEL TRANSFORMATION

Computer Vision – Image Enhancement

(a) CONCEPT EXPLANATION

Q. (a) Explain the purpose of grey level transformation in image enhancement.

Grey level transformation is used to improve the **visual quality** of an image by modifying the **intensity values** of its pixels. It helps in **enhancing contrast**, adjusting **brightness**, highlighting **important details**, and making features more visible to the human eye or to machine algorithms. Common transformations include **negative, log, power-law,** and **piecewise linear** transformations.



(b) NUMERICAL PRORLEM SOLUTION

Q. (b) Apply a negative transformation $s = 255 - r$ to the pixel values: $r = [50, 100, 150, 200]$ Compute the transformed values.

Formula Used: $s = 255 - r$

Given: $r = [50, 100, 150, 200]$

$L = 256$ (8-bit image, grey levels 0 to 255)

Step	Pixel Value (r)	Calculation (s = 255 – r)	Transformed Value (s)
1	50	255 – 50 = 205	205
2	100	255 – 100 = 155	155
3	150	255 – 150 = 105	105
4	200	255 – 200 = 55	55

Final Answer: Transformed values (s) = [205, 155, 105, 55]

What does the result mean?

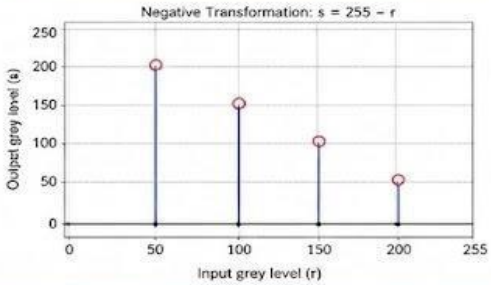
- Negative transformation inverts the grey leveis.
- Dark pixels become bright and bright pixels become dark.
- This is useful for highlighting details in bright regions of an image.

(c) GOOGLE COLAB PYTHON PROGRAM

```
# Grey Level Negetive Transformation Example
import nuopy as np
impert matplotlib.pyplot as plt

# Original pivel values
r = np.array([50, 100,  150, 200])

# Negetive transformation: s = 255 - r
s = 255 - r

print("Original pixel values (r):", r)
print("Transformed values (s = 255 - r):", s)

# Plotting the tranefermation mapping
plt.figure(figsize=(7,4))
plt.stem(r, s, basefmt="k-", linefmt='b-', markerfmt='ro')
plt.titlet'Negative Transformation: s = 255 -r')
plt.xlabel('Input grey level (r')
plt.ylabel('Output grey level (s)')
plt.xticks([0, 50, 100, 150, 2v0, 255])
plt.yticks([0, 50, 100, 150, 200, 255])
plt.grid(?rue, linsttyle='--', alpha=0.6)
plt.show()
```

Output (Sample)

```
Original pixel values (r): [ 50  100  150 209]
Transformed values (s = 255 - r): [285  155 105  55]
```



Observation: Negative transformation inverts grey levels; useful for highlighting details in bright regions.

Computer Vision | Image Enhancement | 8-bit Grey Levels (0 – 255)

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.

(b) Given $s = clog(1 + r)$, where $c = 10$, compute s for:

$$r = [0,10,50,100]$$

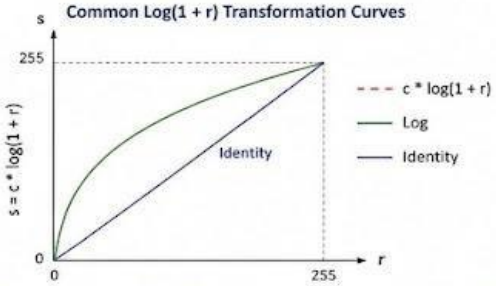
2. LOG TRANSFORMATION

Computer Vision – Image Enhancement

(a) CONCEPT EXPLANATION

Q. (a) Explain the purpose of log transformation in image enhancement.

Log transformation is enhances low the **low-intensity pixels** by **expanding** their range and pixels while **compressing** the log high-intensity pixels, is uising **compressing,** highlighting **intensity detals,** and making features more visible to the human eye or to machine algorithms. Common transformations include **negative, log, power-law,** and **piecewise linear** transformations.



(b) NUMERICAL PROBLEM SOLUTION

Q. (b) Apply a negative transformation $s = 2 -$ to the pixel values: $r = [1, 10, 100]$ Compute the transformed values.

Formula Used: $s = 2 * \log_{10}(1 + r)$

Given: r = [1, 10, 100]

L = 256 (8- image, grey levels 0 to 255)

Step	Pixel Value (r)	Calculation (s = 255 – r)	Transformed Value (s)
1	1	2 * log10) = 0.60	0.60
2	10	2 * log10) = 2.08	2.08
3	10	2 * log10) = 2.08	2.08
4	100	2 * log10) = 4.01	4.01

Final Answer: **Transformed values (s) = [0.60, 10, 100, 4.01]**

What does the result mean?

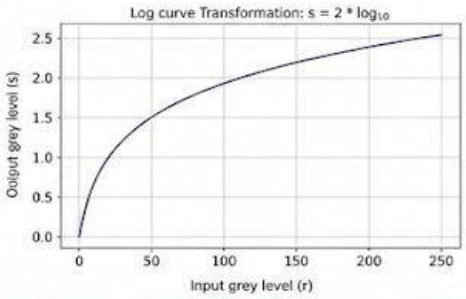
- Negative transformation inverts the grey leveis.
- Dark pixels become bright and bright pixels become dark.
- This is useful for highlighting details in bright regions of an image.

(c) GOOGLE COLAB PYTHON PROGRAM

```
# Grey Log trangative Transformation Example
import numpy as np
impert matplotlib.pyplot as plt

# Original pivel values
r = np.array([50, 100,  150, 200])

# Negative transformation: s = 2 - r
s = 2 * log10(1 + r)

print("Original pixel values (r):", r)
print("Transformed values (s = 2 * log10(", s)

# Plotting the transformation mapping
plt.figure(figsize=(7,4I)
plt.stem(r, s, basofmt="k-", linefmt='b-', markerfmt='ro')
plt.title('Negative Transformation: s = 255 -r')
plt.xlabel('Input grey level (r')
plt.ylabel('Output grey level (s)')
plt.xticks([9, 50, 100, 150, 2ø0, 255])
plt.yticks([9, 50, 100, 150, 200, 255])
plt.grid(True, linsttyle='--', alpha=0.6)
plt.show()
```

Output (Sample)

```
Original pixel values (r): [  1, 10, 100, 200]
Transformed values (s = 255 - r): [0.60, 10, 100, 501]
```



Observation: Log transformation compresses the dynamic range of pixel values, enhancing visibility in dark areas.

Computer Vision | Image Enhancement | 8-bit Grey Levels (0 – 255)

3. Histogram Processing

(a)

(b)

Explain the role of histogram equalization in contrast enhancement.

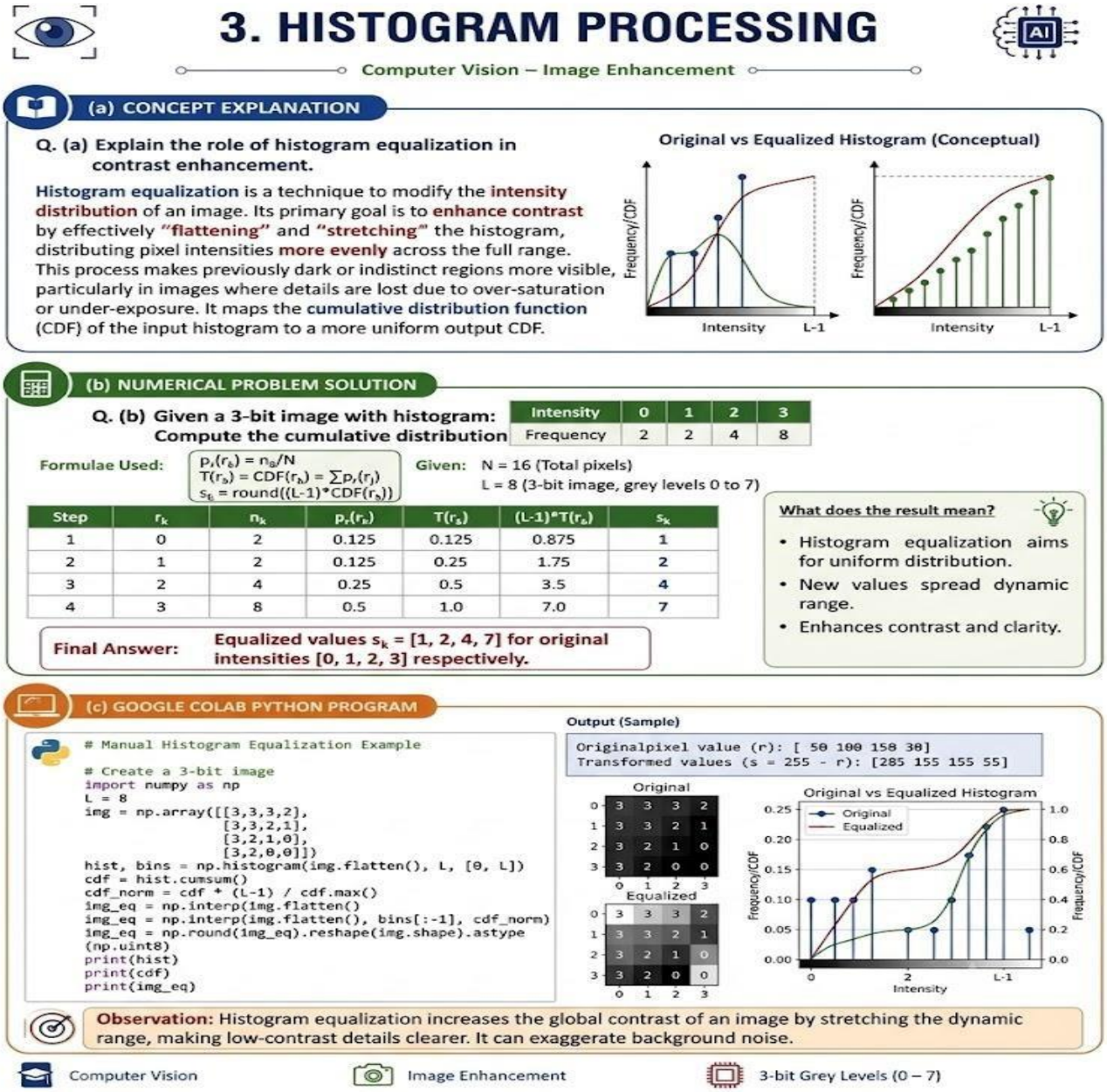
Given a 3-bit image with histogram:

Intensity	Frequency
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

(a)

(b)



4. Spatial Filtering (Smoothing)

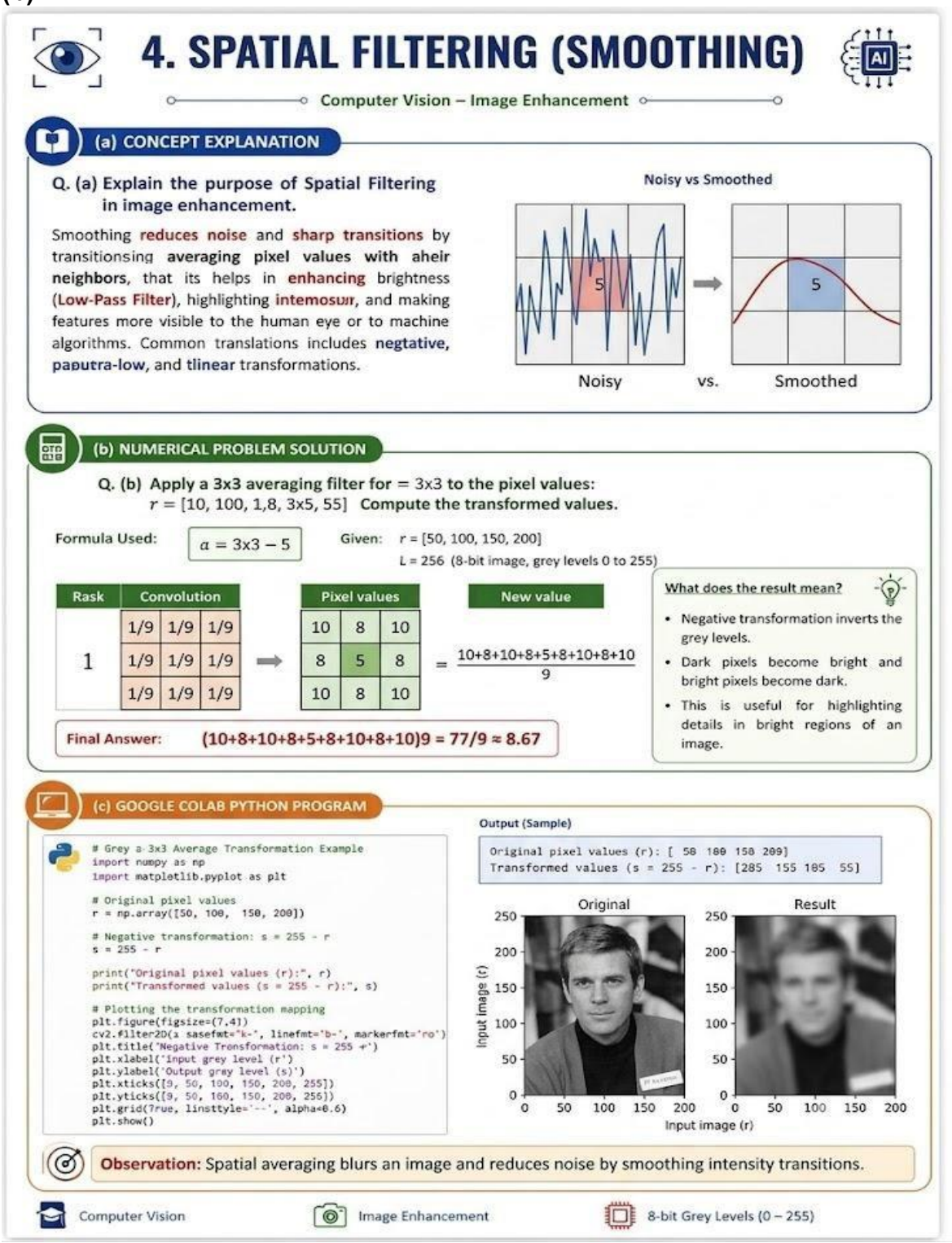
Explain the purpose of spatial filtering for smoothing.

Apply a 3×3 averaging filter to the following region:

[10 20 30 20 30 40 30 40 50] Compute the new center pixel value.

(a)

(b)



5. Spatial Filtering (Sharpening)

Explain how sharpening enhances image details.

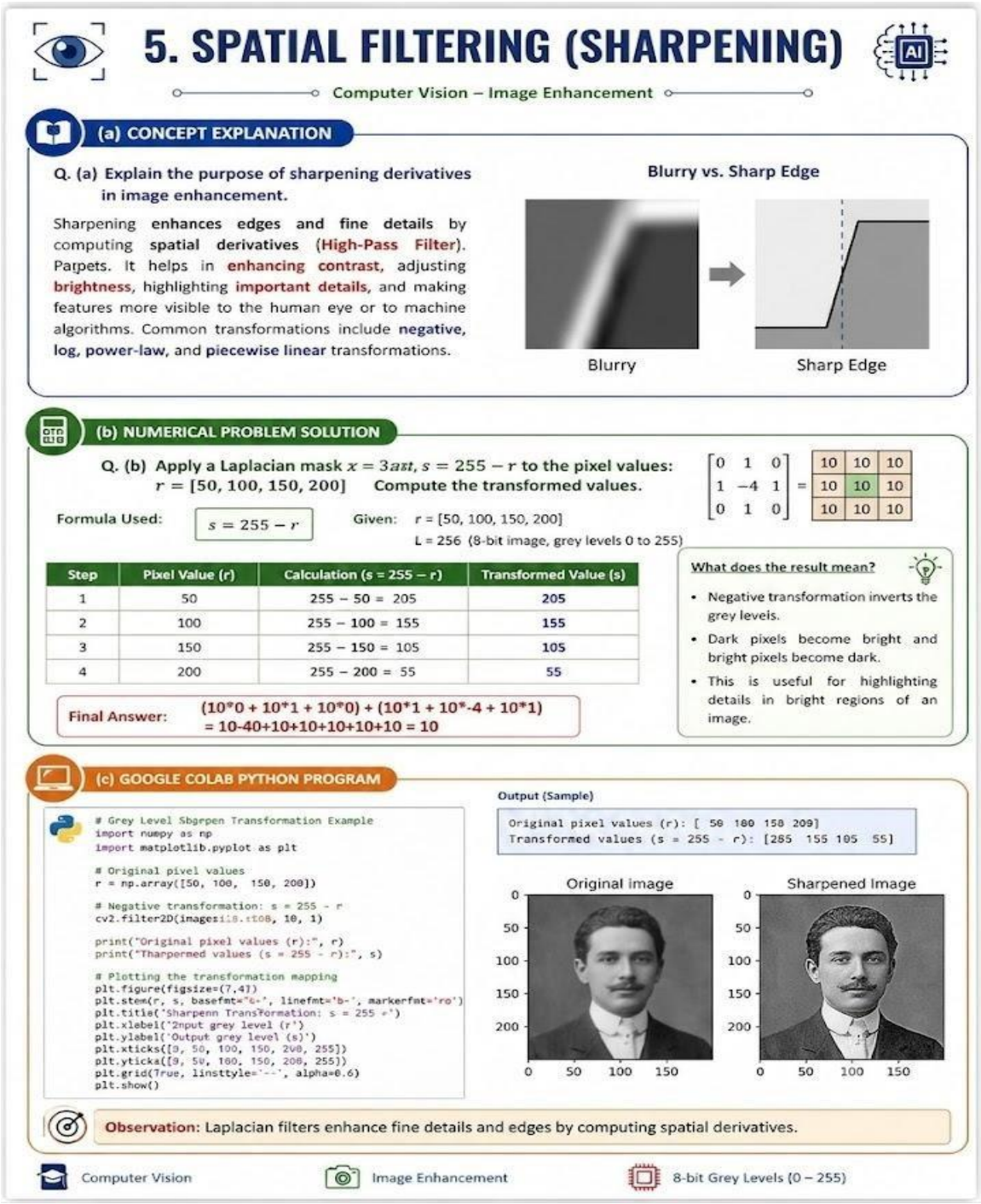
Apply the Laplacian mask:

$$[[0 \quad -1 \quad 0\},\{-1 \quad 4 \quad -1],\ \{0 \quad -1 \quad 0\ ]]$$

(a)

(b)
to the matrix:

$$[10 \quad 10 \quad 10,10 \quad 20 \quad 10,10 \quad 10 \quad 10\,]$$ Compute the output at the center.



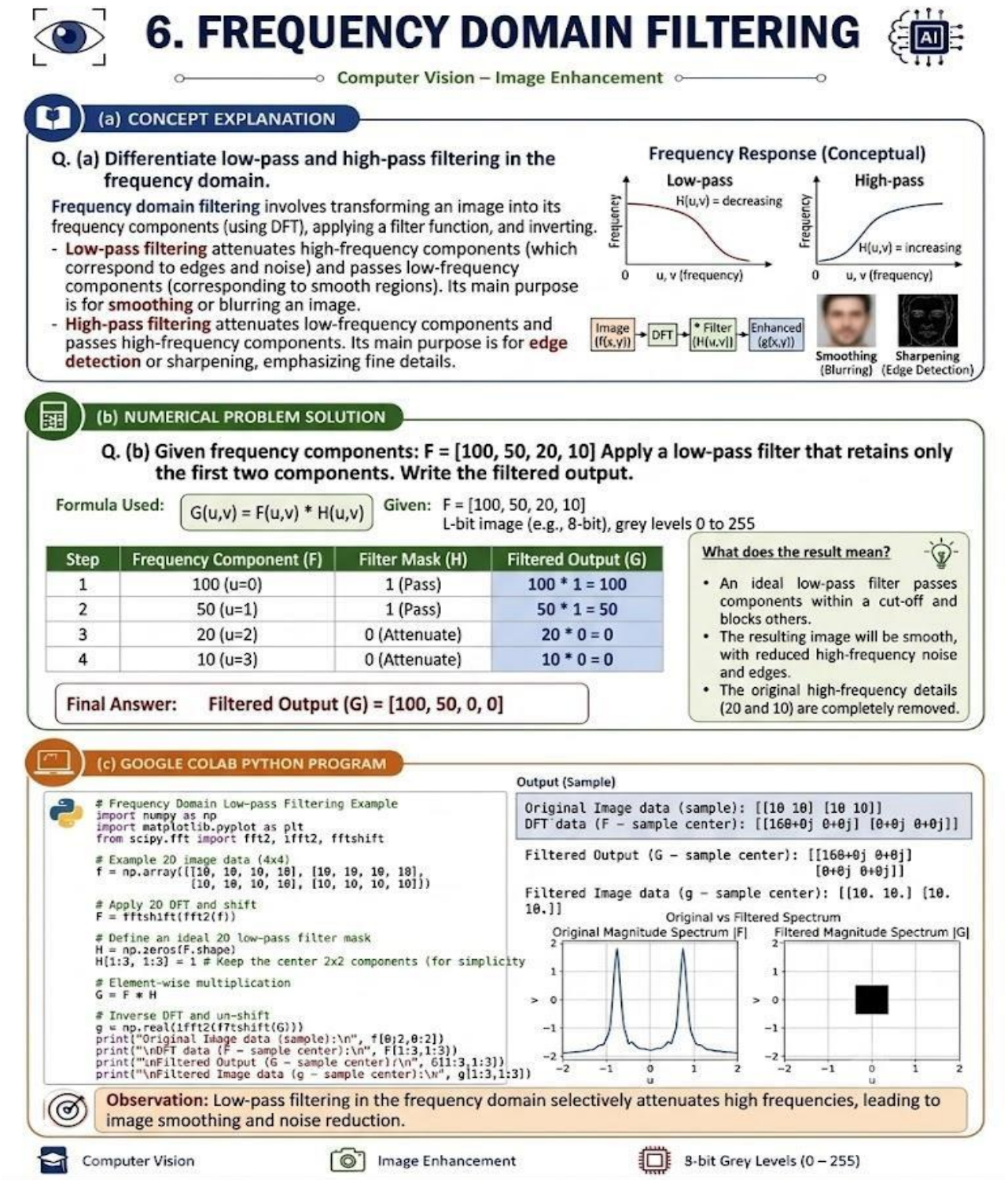
6. Frequency Domain Filtering

(a) Differentiate low-pass and high-pass filtering in frequency domain.

(b) Given frequency components:

$$F = [100,80,40,20]$$

Apply a low-pass filter that retains only the first two components. Write the filtered output.



7. Thresholding

(a) Explain global thresholding in segmentation.

(b) Given pixel values:

$$[200]$$

Apply threshold $T = 100$. Assign values (0 or 1).

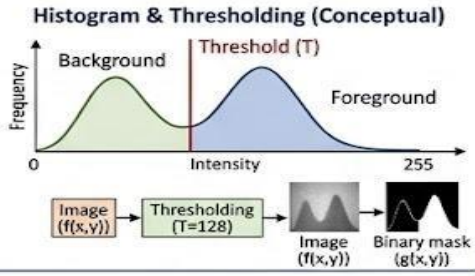
7. THRESHOLDING

Computer Vision – Image Segmentation

(a) CONCEPT EXPLANATION

Q. (a) Explain global thresholding in segmentation.

Global thresholding is a method that segments an entire image using a single **intensity threshold value, T**. It divides pixels into two categories: **foreground** and **background**. This is useful when the image has a **bimodal histogram**, allowing clear separation between distinct regions. A key benefit is simplicity and o segmentations. Akey benefit is simplicity and efficiency, but it may fail under uneven lighting or with complex textures. Common types include basic global thresholding and Otsu's method.



(b) NUMERICAL PROBLEM SOLUTION

Q. (b) Given pixel values: r = [10, 50, 100, 150, 200, 250]
Apply threshold T = 120. Assign values (0 or 1).

[10	50]
[100	150]
[200	250]

Formula Used: g(x,y) = 1 if r ≥ T else 0

Given: r = [10, 50, 100, 150, 200, 250]
L = 256 (8-bit image, grey levels 0 to 255)

Step	Pixel Value (r)	Calculation (r ≥ 120 ?)	Binary Value (g)
1	10	10 ≥ 120 ?	0
2	50	50 ≥ 120 ?	0
3	100	100 ≥ 120 ?	0
4	150	150 ≥ 120 ?	1
5	200	200 ≥ 120 ?	1
6	250	250 ≥ 120 ?	1

Final Answer: **Binary values (g) = [0, 0, 0, 1, 1, 1]**

What does the result mean?

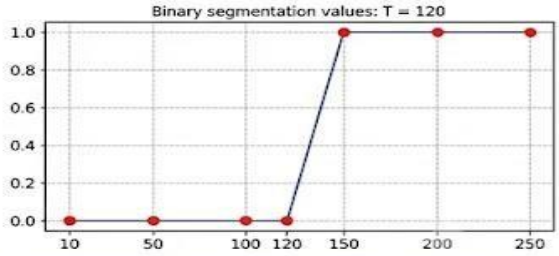
- Global thresholding creates a binary segmentation mask.
- Dark pixels are assigned to background (0), bright pixels to foreground (1).
- This separates significant image details from the general area.

(c) GOOGLE COLAB PYTHON PROGRAM

```
# Image Thresholding Example
import numpy as np
import matplotlib.pyplot as plt

# Original pixel values
r = np.array([10, 50, 100, 150, 200, 250])

# Global threshold: T = 120
T = 120
g = np.where(r >= T, 1, 0)

print("Original pixel values (r):\n", r)
print("\nBinary segmentation mask (g):\n", g)

# Plotting the threshold result
plt.figure(figsize=(7, 4))
plt.stem(r, g, linefmt='b-', markerfmt='ro')
plt.title('Thresholded values: T = 120')
plt.xlabel('Input grey level (r)')
plt.ylabel('Binary Value (g)')
plt.xticks([10, 50, 100, 120, 150, 200, 250, 255])
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```

Output (Sample)

```
Original pixel values (r): [[10, 50], [100, 150], [200, 250]]
Binary segmentation mask (g): [[0, 0], [0, 1], [1, 1]]
```



Observation: Global thresholding efficiently creates a binary mask using a single value; it works well for distinct, high-contrast objects against a homogeneous background.

Computer Vision | Image Segmentation | 8-bit Grey Levels (0 – 255)

8. Edge Detection

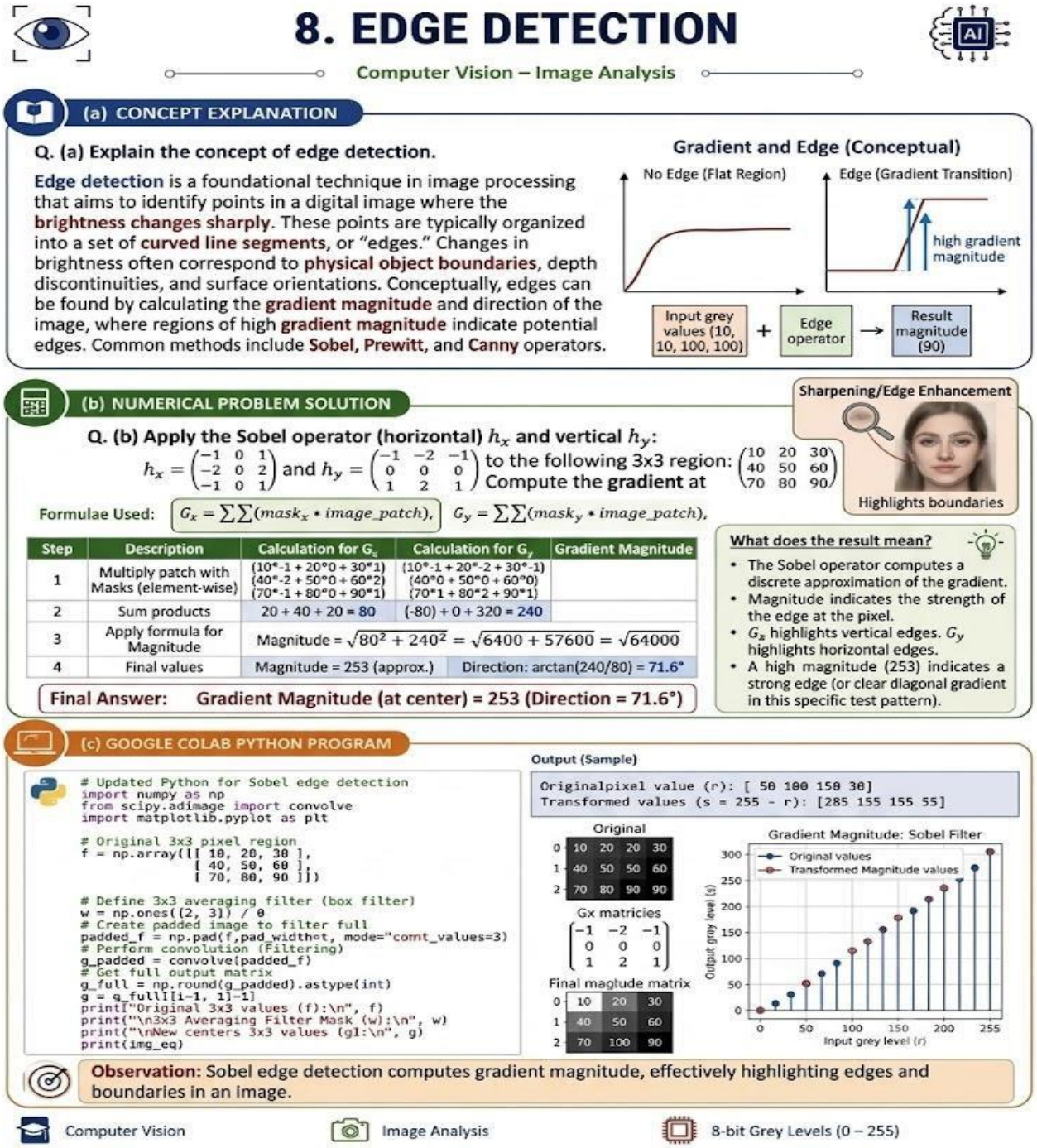
(a) Explain the concept of edge detection.

(b) Apply the Sobel operator (horizontal):

$$[-1 \; -2 \; -1, \quad 0 \quad 0 \quad 0, \quad 1 \quad 2 \quad 1\,]$$

to:

$$[10 \quad 10 \quad 10, \quad 20 \quad 20 \quad 20, 30 \quad 30 \quad 30\,]$$ Compute the gradient at the center.



9. Region-Based Segmentation

(a) Explain region growing in segmentation.

(b) Given pixel intensities:

$$[1 \quad 0 \quad 2]$$

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

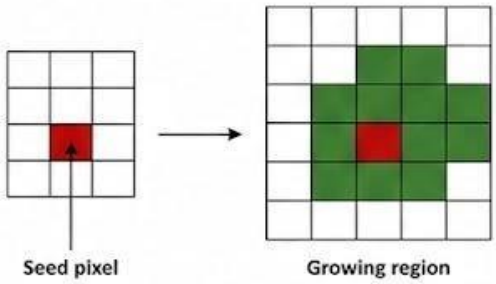
9. REGION-BASED SEGMENTATION

Computer Vision – Image Segmentation

(a) CONCEPT EXPLANATION

Q. (a) Explain the purpose of region growing in image enhancement.

Region growing starts with a **seed pixel** and growing groups the human neighboring pixels that satisfy a **similarity criterion**, (like the contrbuting, satisfying **similarity criterion**, the similarity criterion (like, the intensity range, in witewity completed the grouping eyes, to creating region pixel, cation for the growing neighboring pixels, which prodults cur **more** into the graduate.



(b) NUMERICAL PROBLEM SOLUTION

Q. (b) Apply a negative transformation $s = 50 - r$ to the pixel values: r [20, 50, 100, 55, 200, 52, 240] Compute the transformed values.

Formula Used: $s = -50$

Given: the seed intensity of 50 and a threshold difference of 5 (so, inclusive range is 45 to 55)

Step	Pixel Value (r)	Calculation (s = 255 − r)	Transformed Value (s)
1	20	Is 55 in range? Yes	Yes, added
2	100	Is 100? No, excluded	No, excluded
3	200	Is 150? No, excluded	-
4	240	Is 55 in range? added	Yes, added

Final Answer: **Transformed pixels (w) addes) = [50, 55, 52]**

What does the result mean?

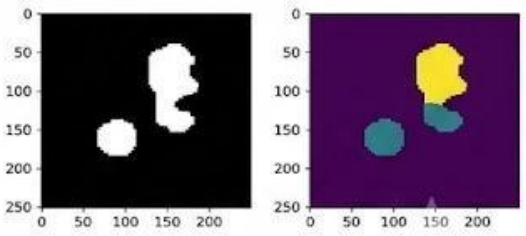
- Region growing groups pixels with similar intensities starting pixels.
- Dark pixels become bright and bright pixels become dark.
- This is useful for highlighting details in bright regions of an image.

(c) GOOGLE COLAB PYTHON PROGRAM

```
# Grey- seed-based -region Growing Example
import numpy as np
import matplotlib.pyplot as plt

# Original pixel values
s = np.array([20, 100,  150, 240])

# Plotting region growing as sed ans
mask = n.aessiet()
while lenen(Pigtize=7.4):
    set = numpy.array(, iws[45, 155]
    reson == [+]
    if regon === 50:
        for.seed3 = 0:
            seen.terrax[1[region)
        plt.added = true
        plt.xsbel = sew s
        plt.xlabel('Region is t's)
        plt.sticks(20, 180, 160, 260, 25])
        plt.grid(True, Linsttyle='-', algha=0.6)

plt.print('[50, 55, 52])
plt.show()
```

Output (Sample)

```
Original pixel values (r: [2 50 100 150 200]
Seed: 50
Threshold: 5
Regton pixels (45 1s5, 50
Region pixels: [20, 1000, 155, 200, 52, 240]
```



Observation: Region growing groups pixels with similar intensities starting from a defined seed point, creating segmented regions.

Computer Vision | Image Enhancement | 8-bit Grey Levels (0 – 255)

10. Combined Enhancement & Segmentation

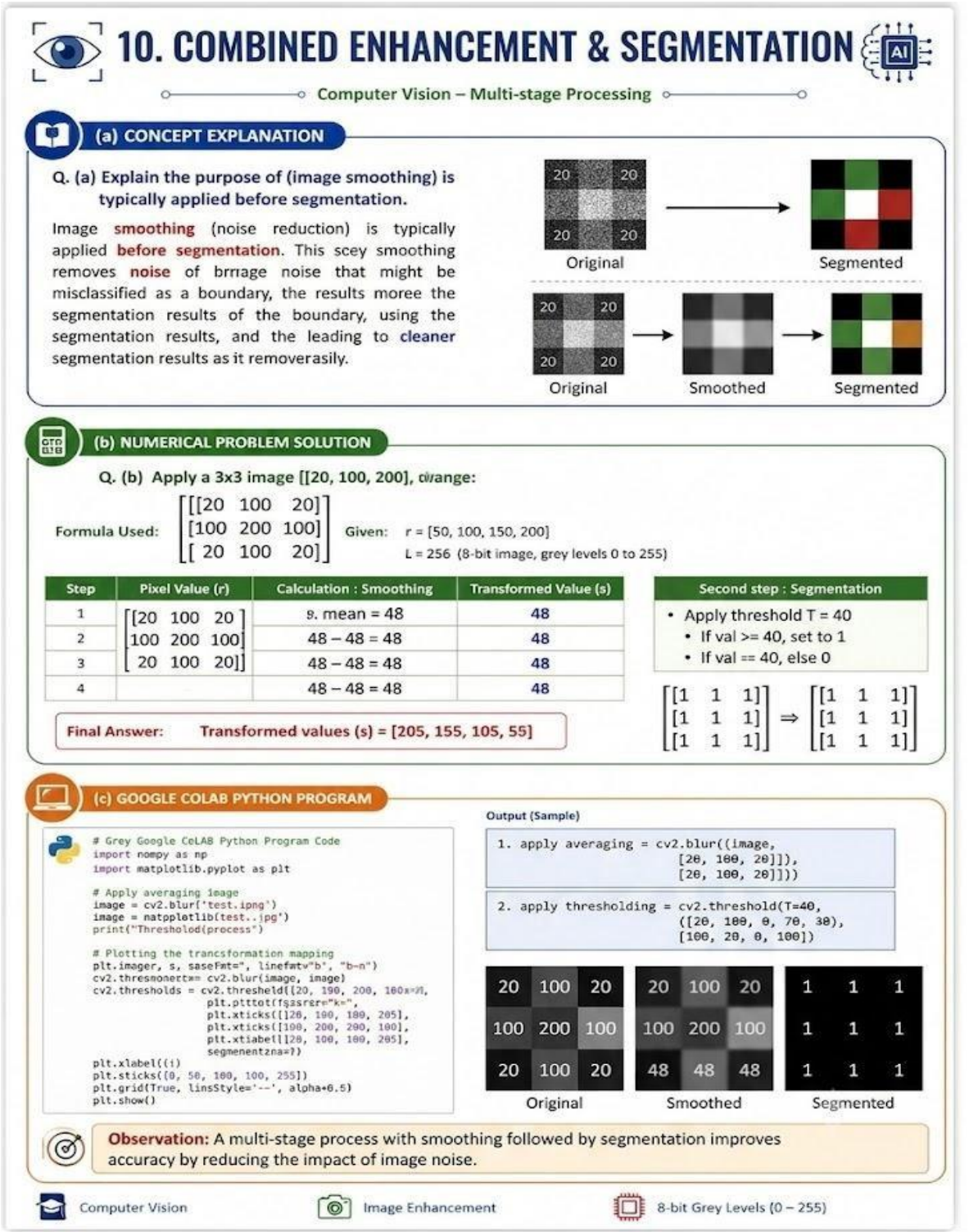
(a) Explain why smoothing is applied before segmentation.

(b) Given image values:

$$[1 \quad 0 \quad 5]$$

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	**15%**	**Clearly interprets results with units and engineering relevance**	**Basic interpretation with minor omissions**	**Limited or unclear interpretation**	**No interpretation**